Dietary fat types and 4-year cognitive change in community-dwelling older women

Objective To relate dietary fat types to cognitive change in healthy community-based elders. Methods Among 6,183 older participants in the Women's Health Study, we related intake of major fatty acids (FAs) (saturated [SFA], mono-unsaturated [MUFA], total poly-unsaturated [PUFA], trans-unsaturated) to late-life cognitive trajectory. Serial cognitive testing, conducted over 4 years, began 5 years post-dietary assessment. Primary outcomes were global cognition (averaging tests of general cognition, verbal memory and semantic fluency) and verbal memory (averaging tests of recall). We used analyses of response profiles and logistic regression to estimate multivariable-adjusted differences in cognitive trajectory and risk of worst cognitive change (worst 10%) by fat intake. Results Higher SFA intake was associated with worse global cognitive (p-linear-trend=0.008) and verbal memory (p-linear-trend=0.01) trajectories. There was a higher risk of worst cognitive change, comparing highest vs. lowest SFA quintiles: the multivariable-adjusted odds ratio (OR) (95% confidence interval, CI) was 1.64 (1.04,2.58) for global cognition and 1.65 (1.04,2.61) for verbal memory. By contrast, higher MUFA intake was related to better global cognitive (p-linear-trend<0.001) and verbal memory (p-linear-trend=0.009) trajectories, and lower OR (95% CI) of worst cognitive change in global cognition (0.52 [0.31,0.88]) and verbal memory (0.56 [0.34,0.94]). Total fat, PUFA, and trans fat intakes were not associated with cognitive trajectory. Interpretation Higher SFA intake was associated with worse global cognitive and verbal memory trajectories, while higher MUFA intake was related to better trajectories. Thus, different consumption levels of the major specific fat types, rather than total fat intake itself, appeared to influence cognitive aging.